

AeroWhip: Aerial Cable Whipping through Iterative Model Refinement

Abstract—A quadrotor can use a cable as a whip to accelerate its free tip toward a target. Model-based planning of this motion must account for both the quadrotor’s tracking response and the resulting cable deformation. We present AeroWhip, a sim-to-real-to-sim framework for aerial cable whipping. The predictor connects a fitted vehicle response, offset attachment geometry, and a reduced differentiable discrete elastic rod (DDER) model. Learned acceleration corrections augment the vehicle and cable dynamics. An offline optimizer inspired by model predictive path integral (MPPI) control plans an initial whipping maneuver. Physical executions refine the vehicle and cable dynamics for command-to-tip prediction and subsequent planning. We apply the updated model to local command correction toward the original planned tip trajectory at fixed times. We evaluate prediction accuracy and the utility of the refined model for physical reference tracking and target approach. **Add quantitative findings after completing the final physical experiments.**

I. INTRODUCTION

A rapid quadrotor maneuver can turn a hanging cable into a whip, directing its free tip toward a target. The vehicle actuates the attached end, and deformation along the cable produces the tip motion. Ropes and cables already provide lightweight tools for suspended-payload transport and aerial grasping [1], [2]. Whipping requires predicting how rapid vehicle commands produce cable deformation and tip motion.

Whipping and dynamic cable manipulation have been studied through model-based optimization with robotic arms [3], movement primitives in a simulated human arm [4], and learning-based control [5]. Aerial whipping adds the challenge of maintaining flight while driving the cable. A quadrotor changes its attitude to accelerate horizontally, so both translation and rotation move an offset attachment. Tracking errors during rapid flight further alter the executed motion [6]. Predicting the attachment trajectory therefore requires a model of the vehicle’s response to commands.

Aerial cable research provides foundations for this prediction. Suspended-payload studies address trajectory generation [1], flexible-cable dynamics [7], agile flight [8], and adaptation to changes in flight dynamics [9]. Work with free-ended cables has demonstrated parameter identification and dynamic shape control using a single quadrotor [10], while related research combines identification from measured endpoint motion with cable manipulation using multiple UAVs [11]. For whipping, the predictor must connect the vehicle’s tracking response to cable deformation and the resulting free-tip trajectory.

Rod mechanics provides a basis for predicting that deformation. Discrete elastic rods represent distributed bending and twisting [12], and DEFORM combines differentiable

rod mechanics with learning for cable prediction, tracking, and manipulation [13]. Related studies investigate physical parameter identification [14] and structured dynamics learning from limited measurements [15]. These approaches motivate a cable predictor that retains rod mechanics and learns corrections from measured motion.

Execution measurements can improve subsequent behavior by updating actions or the model used to plan them. Iterative Residual Policy refines actions from previous executions [16]. Suresh and Atkeson [17] use task-level iterative learning control for dynamic rope manipulation, mapping measured task errors to command updates through a local inverse model of the robot and rope. Model-based iterative learning has also improved quadrotor tracking by updating feedforward commands across repetitions [18]. Real2Sim2Real uses cable observations to improve simulation and subsequent casting actions [19], while Wiggle and Go identifies rope properties through a probing motion before planning dynamic tasks [20]. Sampling-based planning has also been combined with repeated fitting of learned dynamics [21]. AeroWhip refines the vehicle and cable dynamics from measured motion, yielding an updated command-to-tip model for prediction and planning. Fixed-reference command correction provides one application of this model.

We present *AeroWhip*, a sim-to-real-to-sim framework for aerial cable whipping through iterative model refinement. Its predictor combines vehicle response, offset attachment geometry, and reduced rod mechanics with learned acceleration corrections. An offline optimizer inspired by model predictive path integral (MPPI) control [22] balances cable-shape agreement, directed tip motion, and target approach under motion constraints. Physical executions refine both the vehicle and cable models. These updates support command-to-tip prediction and subsequent planning. Here, local command correction uses the updated predictor to reproduce the original planned tip motion at fixed times. We evaluate prediction accuracy and whether model-guided correction improves physical reference tracking and target approach.

II. PROBLEM STATEMENT

A quadrotor carries a cable of length $L > 0$, attached at one end and free at the other. The attachment point is fixed relative to the vehicle, and the cable is modeled as freely pivoting there. We assume uniform material properties, negligible axial extension, a naturally straight and untwisted undeformed configuration, and isotropic bending stiffness. Planning starts from nominal hover with the cable hanging at rest.

The task is to bring the free tip close to a stationary target $\mathbf{p}_{\text{tar}} \in \mathbb{R}^3$ with rapid motion along a prescribed horizontal unit direction $\mathbf{e} \in \mathbb{R}^3$. The strike direction specifies the desired tip motion at the target, independently of the target's position relative to launch. Let $t = 0$ denote maneuver onset and $T > 0$ the targeting horizon. The planner generates desired quadrotor position, velocity, acceleration, and yaw trajectories, which are fixed before execution and tracked by the existing flight controller.

A. Whipping Motion

Figure 1a illustrates the whipping motion considered in AeroWhip. A forward and upward pull is followed by retraction, during which a bend travels toward the free end as the cable straightens and the tip approaches the target. This schematic and travelling-deformation descriptions of whipping [23] motivate characterizing the motion through both cable-shape evolution and tip kinematics.

Let $\mathbf{r}(s, t) \in \mathbb{R}^3$ denote the world-frame position of the cable material point at unstretched distance sL from the attachment, with $s \in [0, 1]$. At a fixed time t , varying s traces the centerline $\mathbf{r}(\cdot, t)$; fixing s follows the same material point throughout the maneuver. The attachment and tip positions are $\mathbf{p}_a(t) = \mathbf{r}(0, t)$ and $\mathbf{r}_{\text{tip}}(t) = \mathbf{r}(1, t)$, respectively, with tip velocity $\mathbf{v}_{\text{tip}}(t) = \dot{\mathbf{r}}_{\text{tip}}(t)$.

We compare predicted and reference shapes at the same $M \geq 3$ material coordinates $s_j = j/(M-1)$, $j = 0, \dots, M-1$. These samples are equally spaced along the unstretched cable and need not coincide with measurement markers. The unit direction between neighboring samples is

$$\mathbf{d}_j(t) = \frac{\mathbf{r}(s_{j+1}, t) - \mathbf{r}(s_j, t)}{\|\mathbf{r}(s_{j+1}, t) - \mathbf{r}(s_j, t)\|_2}, \quad (1)$$

for $j = 0, \dots, M-2$, assuming distinct neighboring positions. Sampled positions describe the centerline, and changes between neighboring directions describe bending. Their histories support the planner's reference-shape comparison.

Physical target approach is measured by the minimum three-dimensional tip-to-target distance:

$$d_{\min} = \min_{t \in [0, T]} \|\mathbf{r}_{\text{tip}}(t) - \mathbf{p}_{\text{tar}}\|_2. \quad (2)$$

For physical evaluation, we substitute the measured trajectory $\mathbf{r}_{\text{tip}}^{\text{meas}}$ into (2). Piecewise-linear interpolation within valid tracking intervals estimates the minimum; tracking gaps are excluded and reported. Tip velocity describes approach speed and alignment with $\mathbf{e}$. Initial planning uses tip kinematics and shape agreement; subsequent correction tracks the resulting planned tip trajectory (Section III-B).

B. DDER Cable Model

We represent the cable with a reduced differentiable discrete elastic rod (DDER) model based on the rod mechanics of Bergou et al. [12].

As illustrated in Figure 1b, we discretize the centerline using $N \geq 3$ simulation nodes with positions $\mathbf{r}_i(t) \in \mathbb{R}^3$ and velocities $\mathbf{v}_i(t) \in \mathbb{R}^3$, $i = 0, \dots, N-1$. Nodes follow fixed material locations. The vectors $\mathbf{r}(t), \mathbf{v}(t) \in \mathbb{R}^{3N}$ stack

their positions and velocities, respectively. Node 0 follows the prescribed attachment, $\mathbf{r}_0 = \mathbf{p}_a$, and node $N-1$ is the free tip, $\mathbf{r}_{N-1} = \mathbf{r}_{\text{tip}}$.

Segment i connects nodes i and $i+1$ and has fixed length $\ell_i > 0$. Inextensibility imposes

$$\|\mathbf{r}_{i+1} - \mathbf{r}_i\|_2 = \ell_i, \quad \sum_{i=0}^{N-2} \ell_i = L.$$

Linear interpolation between nodes defines the model centerline $\mathbf{r}(s, t)$ and the material samples for shape comparison. The simulation-node count N and shape-sample count M need not be equal.

Material twist is the relative rotation of cable cross-sections about the centerline. We assume these orientations remain at elastic equilibrium, with no imposed end twist or twisting moment. For the naturally straight, untwisted, isotropic rod, this gives zero material twist [12]. We omit independent twist coordinates and torsional energy while retaining three-dimensional centerline dynamics.

Bending energy penalizes changes in direction between adjacent simulation segments. Define the unit segment tangent

$$\mathbf{t}_i = \frac{\mathbf{r}_{i+1} - \mathbf{r}_i}{\|\mathbf{r}_{i+1} - \mathbf{r}_i\|_2}.$$

Tangents $\mathbf{t}_i$ describe model segments; directions $\mathbf{d}_j$ in (1) describe shape-sampling intervals. For each interior node $i = 1, \dots, N-2$, let $\bar{\ell}_i = (\ell_{i-1} + \ell_i)/2$ be the average rest length of its adjacent segments. The discrete curvature binormal and bending energy are

$$\boldsymbol{\kappa}_i^b = \frac{2 \mathbf{t}_{i-1} \times \mathbf{t}_i}{1 + \mathbf{t}_{i-1}^\top \mathbf{t}_i}, \quad E_b(\mathbf{r}) = \frac{EI}{2} \sum_{i=1}^{N-2} \frac{\|\boldsymbol{\kappa}_i^b\|_2^2}{\bar{\ell}_i}, \quad (3)$$

where $EI > 0$ is the bending stiffness. The dimensionless curvature binormal measures turning between adjacent segments and, when nonzero, is normal to their plane. It vanishes for aligned tangents and is singular for antiparallel tangents.

The ideal bending force is $\mathbf{f}_b = -\nabla_{\mathbf{r}} E_b$. Numerical evaluation uses a positive denominator floor near antiparallel tangents. Time integration, constraint enforcement, and learned acceleration corrections are described in Section III-A.

Admissible commands satisfy prescribed motion bounds and pass simulation checks on vehicle altitude, cable ground clearance, and numerical validity using the current predictor. Section III-B specifies these conditions.

Problem (Aerial Cable Whipping). Given an initial hovering configuration, target $\mathbf{p}_{\text{tar}}$, strike direction $\mathbf{e}$, horizon T , and a reference shape history, generate admissible quadrotor commands that bring the tip close to the target through the desired whipping motion. Use physical executions to refine the predictor and correct commands toward the selected motion. The problem comprises three connected subproblems.

Subproblem 1 (Command-to-Tip Prediction). Predict cable motion from desired quadrotor commands, accounting for vehicle response, attachment geometry, and cable dynamics.

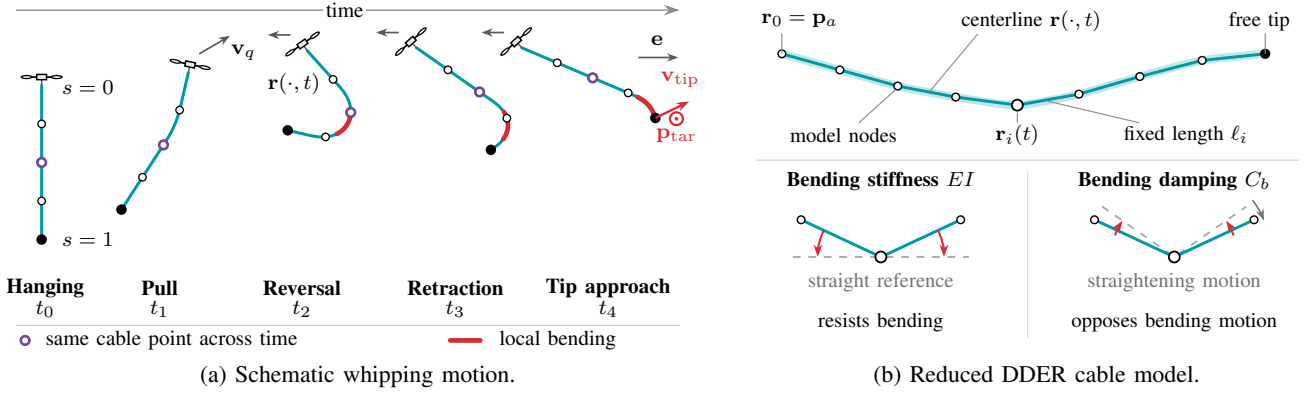


Fig. 1: Whipping motion and cable model. (a) Schematic whipping sequence; purple follows one cable point, while red highlights bending. (b) Node positions $\mathbf{r}_i(t)$ discretize the centerline $\mathbf{r}(\cdot, t)$. Red arrows illustrate the responses of bending stiffness EI and bending damping C_b .

Subproblem 2 (Offline Trajectory Optimization). Plan an admissible whipping command and, after model refinement, correct it toward the original predicted tip trajectory.

Subproblem 3 (Model Refinement). Update the predictor using recorded commands and measured vehicle and cable motion to support command correction and subsequent planning.

III. AEROWHIP FRAMEWORK

AeroWhip combines command-to-tip prediction, offline trajectory optimization, and model refinement. An initial MPPI plan defines the desired whipping motion. Physical executions supply measurements for refining the vehicle and cable models, which then guide local command correction toward that motion. The refined predictor also remains available for subsequent simulation and planning. Below, we describe the predictor, the two command optimization stages, and the model-fitting procedure.

A. Quadrotor and Cable Prediction Model

The predictor maps commands to vehicle motion, then uses the resulting attachment trajectory to drive the cable model. The fitted vehicle response approximates cable-loading effects under the observed motion conditions; cable configuration and reaction forces are not explicit inputs. Vehicle and cable states below denote predictions unless marked as measured.

The inputs are desired position $\mathbf{p}_d$, velocity $\mathbf{v}_d$, acceleration $\mathbf{a}_d$, and yaw ψ_d , held constant between command updates. Let $\mathbf{p}_q$ and $\mathbf{v}_q$ denote the position and velocity of the vehicle's tracked origin, and let $\mathbf{R} \in \text{SO}(3)$ map vectors from the vehicle's body-fixed tracking frame to the world frame. We approximate the closed-loop translational response by

$$\begin{aligned} \mathbf{a}_{\text{nom}}(t) = & \mathbf{K}_p[\mathbf{p}_d(t - \tau_d) - \mathbf{p}_q(t)] \\ & + \mathbf{K}_v[\mathbf{v}_d(t - \tau_d) - \mathbf{v}_q(t)] \\ & + \mathbf{K}_a\mathbf{a}_d(t - \tau_d) + \mathbf{b}_q, \end{aligned} \quad (4)$$

where $\tau_d \geq 0$ is an effective delay and $\mathbf{b}_q$ is a constant acceleration bias within each rollout. The fitted diagonal

gains $\mathbf{K}_p$, $\mathbf{K}_v$, and $\mathbf{K}_a$ each share a horizontal coefficient and have a separate vertical coefficient. Translation satisfies $\dot{\mathbf{p}}_q = \mathbf{v}_q$ and $\dot{\mathbf{v}}_q = \mathbf{a}_{\text{nom}} + \Delta\mathbf{a}_q$, where $\Delta\mathbf{a}_q$ is a learned acceleration correction.

Vehicle attitude determines the rotation of the offset attachment. We construct a commanded rotation $\mathbf{R}_c$ with its third axis parallel to $\mathbf{S}\mathbf{a}_{\text{nom}} + g\mathbf{e}_3$, where $g > 0$ is gravitational acceleration, $\mathbf{e}_3 = (0, 0, 1)^\top$ is the upward world direction, and $\mathbf{S}$ is a fitted positive diagonal matrix with dimensionless horizontal and vertical scales. The first axis is the normalized projection of the delayed yaw heading $\psi_d(t - \tau_d)$ onto the plane normal to the third axis; the second completes a right-handed frame. Zero specific force or a parallel heading makes this construction invalid.

An effective alignment $\mathbf{Q} \in \text{SO}(3)$, fixed during each prediction, maps tracking-frame coordinates to the vehicle-axis convention used for $\mathbf{R}_c$, giving the desired orientation of the tracking frame, $\mathbf{R}_d = \mathbf{R}_c\mathbf{Q}$. Let $\mathbf{e}_R$ be the principal rotation vector of $\mathbf{R}^\top\mathbf{R}_d$, expressed in the tracking frame. The attitude response is

$$\dot{\mathbf{R}} = \mathbf{R}[\boldsymbol{\omega}]_\times, \quad \dot{\boldsymbol{\omega}} = \frac{\mathbf{e}_R}{\tau_R^2} - \frac{2\boldsymbol{\omega}}{\tau_R}, \quad (5)$$

where $\boldsymbol{\omega}$ is tracking-frame angular velocity, $\tau_R > 0$ is a fitted response timescale, and $[\boldsymbol{\omega}]_\times$ denotes the cross-product matrix. The commanded orientation uses nominal acceleration; $\Delta\mathbf{a}_q$ affects later attitude predictions through the evolving position and velocity errors.

For a fixed tracking-frame offset $\boldsymbol{\rho}_a$ from the tracked origin to the attachment, the attachment position is

$$\mathbf{p}_a = \mathbf{p}_q + \mathbf{R}\boldsymbol{\rho}_a. \quad (6)$$

This trajectory prescribes the moving cable root, $\mathbf{r}_0 = \mathbf{p}_a$, with root velocity obtained from finite differences of the interpolated positions.

For each free node $i = 1, \dots, N - 1$,

$$\begin{aligned} m_i\dot{\mathbf{v}}_i = & \mathbf{f}_{b,i} + \mathbf{f}_{\text{bd},i} + \mathbf{f}_{\text{con},i} \\ & - m_i\gamma_c\mathbf{v}_i - m_i g\mathbf{e}_3 + m_i\Delta\mathbf{a}_{c,i}, \end{aligned} \quad (7)$$

with $\dot{\mathbf{r}}_i = \mathbf{v}_i$. Here m_i is the node mass, $\mathbf{f}_{b,i}$ is the bending force from (3), and $\mathbf{f}_{bd,i}$ is the bending-damping force. The reaction $\mathbf{f}_{\text{con},i}$ enforces fixed segment lengths, $\gamma_c \geq 0$ is the linear-drag rate, and $\Delta \mathbf{a}_{c,i}$ is a learned acceleration correction.

Bending damping uses a regularized curvature rate that approximately removes the contribution of local rigid rotation. At each interior node $i = 1, \dots, N-2$, define

$$\mathbf{n}_i = \mathbf{t}_{i-1} \times \mathbf{t}_i, \quad c_i = \mathbf{t}_{i-1}^\top \mathbf{t}_i, \quad \alpha_i = \frac{\varepsilon}{\|\mathbf{n}_i\|_2^2 + \varepsilon},$$

where $\varepsilon > 0$ is a fixed dimensionless regularization parameter. Using the curvature binormal $\boldsymbol{\kappa}_i^b$ from (3), the local angular-velocity estimate and regularized bending rate operator are

$$\begin{aligned} \boldsymbol{\omega}_i^{\text{loc}} &= \frac{1}{2}(\mathbf{I} - \mathbf{t}_{i-1}\mathbf{t}_{i-1}^\top)(\mathbf{t}_{i-1} \times \dot{\mathbf{t}}_{i-1} + \mathbf{t}_i \times \dot{\mathbf{t}}_i), \\ \dot{\boldsymbol{\kappa}}_{i,\text{rel}}^b &= -\frac{2\mathbf{n}_i\dot{c}_i}{(1+c_i)(\|\mathbf{n}_i\|_2^2 + \varepsilon)} \\ &\quad + \alpha_i(\boldsymbol{\kappa}_i^b - \boldsymbol{\omega}_i^{\text{loc}} \times \boldsymbol{\kappa}_i^b). \end{aligned} \quad (8)$$

Here $\mathbf{I}$ is the 3×3 identity matrix, and dots denote time derivatives induced by node velocities. The regularization keeps the rate finite when adjacent tangents point in the same direction.

The dissipation function and stacked bending-damping force are

$$\mathcal{D}_b = \frac{C_b}{2} \sum_{i=1}^{N-2} \frac{\|\dot{\boldsymbol{\kappa}}_{i,\text{rel}}^b\|_2^2}{\bar{\ell}_i}, \quad \mathbf{f}_{bd} = -\nabla_{\mathbf{v}} \mathcal{D}_b, \quad (9)$$

where $C_b \geq 0$ is the bending-damping coefficient. The cable state is advanced with implicit bending damping and mass-weighted position and velocity projections for the inextensibility constraints.

Two multilayer perceptrons provide the learned acceleration corrections. The vehicle network takes the position and velocity tracking errors in (4), delayed desired acceleration, predicted velocity, and bias $\mathbf{b}_q$, and outputs $\Delta \mathbf{a}_q$. The cable network takes all node positions and velocities relative to the attachment, $\mathbf{r}_i - \mathbf{r}_0$ and $\mathbf{v}_i - \mathbf{v}_0$, together with attachment velocity $\mathbf{v}_0$. It jointly outputs $\Delta \mathbf{a}_{c,i}$ for all free nodes; the prescribed root receives no correction.

All vector inputs and outputs use world-frame components. Inputs receive fixed scaling, and a scaled hyperbolic tangent bounds each correction component by $a_{\text{max}} > 0$. A motion-dependent factor suppresses the vehicle correction at rest and increases toward one with the magnitudes of predicted velocity, delayed desired velocity, and delayed desired acceleration. Network weights remain fixed during each rollout and are fitted as described in Section III-C.

Planning and preflight forecasts start from nominal hover, with the tracking frame and commanded heading aligned with the world axes: $\mathbf{R}(0) = \mathbf{Q} = \mathbf{I}$ and $\mathbf{b}_q = \mathbf{0}$. Initial vehicle, cable, and angular velocities are zero, and the cable hangs straight. Before maneuver onset, commands hold the initial desired position with zero desired velocity and acceleration. Fitting and postflight model comparisons initialize the state, bias, and alignment from pre-maneuver

measurements. These comparisons are distinct from the saved preflight forecasts.

B. Offline Trajectory Optimization

Initial planning uses MPPI-inspired sampling to select a whipping motion. After model refinement, local optimization corrects commands toward its predicted tip trajectory. Both stages use the same B-spline representation and hold model parameters fixed.

We represent the desired vehicle position by a smooth command spline using a fixed clamped quintic B-spline basis:

$$\mathbf{p}_s(t; \boldsymbol{\theta}) = \sum_{j=0}^{n_c-1} B_j(t) \mathbf{p}_j^{\text{ctrl}}, \quad t \in [0, T]. \quad (10)$$

Here B_j are the basis functions and $\mathbf{p}_j^{\text{ctrl}} \in \mathbb{R}^3$ are the $n_c \geq 6$ control points. We fix $\mathbf{p}_0^{\text{ctrl}} = \mathbf{p}_1^{\text{ctrl}} = \mathbf{p}_2^{\text{ctrl}} = \mathbf{p}_q(0)$, giving the initial position and zero initial velocity and acceleration. The remaining control points form $\boldsymbol{\theta} \in \mathbb{R}^{3(n_c-3)}$. Desired position, velocity, and acceleration are sampled from $\mathbf{p}_s$, $\dot{\mathbf{p}}_s$, and $\ddot{\mathbf{p}}_s$, respectively, and held between command updates. Desired yaw remains constant. The spline end is free to support follow-through.

Initial planning fits the spline to a selected seed command under jerk bounds. Candidate rollouts must satisfy bounds on spline jerk, sampled command speed and height, and the magnitude, tilt, and vertical component of $\mathbf{a}_d + g\mathbf{e}_3$. Admissibility also requires predicted vehicle-height limits, cable ground clearance, and numerical validity.

For initial whip design, the optimizer maximizes

$$\begin{aligned} J(\boldsymbol{\theta}) &= w_{\text{whip}} S_{\text{tar}} S_{\text{whip}} + w_{\text{enc}} S_{\text{enc}} + w_{\text{ext}} S_{\text{ext}} \\ &\quad - w_{\text{miss}}(1 - S_{\text{tar}}) - C_{\text{motion}} - C_{\text{exit}}, \end{aligned} \quad (11)$$

where S_{tar} measures target proximity and S_{whip} measures reference-shape agreement. The scores S_{enc} and S_{ext} favor vehicle retraction and directed tip extension, while C_{motion} and C_{exit} penalize vehicle motion and recovery handover conditions. The four score weights are positive; all terms are defined below and evaluated on the candidate rollout.

Target approach and directed motion. A sphere of radius $r_{\text{tar}} > 0$ centered at $\mathbf{p}_{\text{tar}}$ defines the planning tolerance region. The encounter time t_e is the first entry by any cable node over $[0, T]$; if none enters, it is the time of closest tip approach. Entry is evaluated along interpolated node trajectories; the tolerance sphere does not exert collision forces. With $d_e = \|\mathbf{r}_{\text{tip}}(t_e) - \mathbf{p}_{\text{tar}}\|_2$, the proximity score is $S_{\text{tar}} = \exp[-(d_e/\sigma_p)^2]$ for $\sigma_p > 0$. This encounter-based distance differs from the physical minimum-distance metric in (2).

At t_e , let $v_{\text{tip},\parallel} = \mathbf{e}^\top \mathbf{v}_{\text{tip}}$ and $v_{q,\parallel} = \mathbf{e}^\top \mathbf{v}_q$ denote the forward tip and vehicle velocities. Let b_e be the vehicle's retraction from its greatest preceding forward position, and let $\ell_e = \mathbf{e}^\top (\mathbf{r}_{\text{tip}} - \mathbf{p}_a)$ be the forward attachment-to-tip extension. The indicator χ is one when no node enters the sphere or when the tip enters strictly before every other node, and zero otherwise. Define $\text{clip}(z) = \min(1, \max(0, z))$

and $h(z) = h_{\min} + (1 - h_{\min}) \text{clip}(z)$, with a fixed floor $0 < h_{\min} < 1$. Then

$$\begin{aligned} S_{\text{enc}} &= S_{\text{tar}} \chi(v_{\text{tip},\parallel}/v_{\text{tip},s}) h(-v_{q,\parallel}/v_{q,s}) h(b_e/b_s), \\ S_{\text{ext}} &= S_{\text{tar}} \chi \text{clip}(v_{\text{tip},\parallel}/v_{\text{tip},s}) \text{clip}(\ell_e/L) \\ &\quad \times \text{clip}(v_{\text{tip},\parallel}/\|\mathbf{v}_{\text{tip}}\|_2)^2. \end{aligned} \quad (12)$$

The positive scales $v_{\text{tip},s}$, $v_{q,s}$, and b_s set the saturation levels. The alignment factor favors the strike direction and is defined as zero at zero tip speed. The positive floor in h makes vehicle reversal a preference rather than a requirement.

Whipping-shape agreement. An earlier simulated whip supplies the reference shape history, denoted by the superscript ref. It was generated with a different development model and guides initial design alongside the seed command.

The normalized samples $\bar{\mathbf{r}}_j(t) = [\mathbf{r}(s_j, t) - \mathbf{p}_a(t)]/L$ remove attachment translation and cable-length scale while retaining world orientation. Apply the same normalization to the reference using its own attachment position and cable length. Shape comparison uses these samples and the directions $\mathbf{d}_j(t)$ in (1).

We align encounter times and allow limited intermediate retiming. For $K \geq 2$ phases $u_k = k/(K-1)$, $k = 0, \dots, K-1$, define paired times $t_k^{\text{ref}} = t_e^{\text{ref}} u_k$ and $t_k^\gamma = t_e u_k^\gamma$. Here $t_e^{\text{ref}} > 0$ is the reference encounter time, and γ belongs to a fixed finite set Γ of positive values containing 1. The shape score is

$$\begin{aligned} S_{\text{whip}} &= \max_{\gamma \in \Gamma} \exp \left[- \frac{\langle \|\bar{\mathbf{r}}_j(t_k^\gamma) - \bar{\mathbf{r}}_j^{\text{ref}}(t_k^{\text{ref}})\|_2^2 \rangle_{j,k}}{\sigma_r^2} \right. \\ &\quad \left. - \frac{\langle \|\mathbf{d}_j(t_k^\gamma) - \mathbf{d}_j^{\text{ref}}(t_k^{\text{ref}})\|_2^2 \rangle_{j,k}}{\sigma_d^2} - \beta |\log \gamma| \right]. \end{aligned} \quad (13)$$

The brackets average over $k = 0, \dots, K-1$ and $j = 0, \dots, M-1$ for positions or $j = 0, \dots, M-2$ for directions, pairing the same material locations. The positive scales σ_r, σ_d are dimensionless, and $\beta \geq 0$ penalizes departure from uniform phase sampling. We set $S_{\text{whip}} = 0$ when $t_e = 0$. This score compares deformation histories; the horizon, directed-speed rewards, and command constraints govern physical timing.

The motion cost C_{motion} accumulates time-weighted squared vehicle displacements from launch and command-spline jerk. Jerk is normalized by its componentwise bounds. Displacement penalties use positive forward motion and lateral motion along $\mathbf{e}_3 \times \mathbf{e}$. Accumulation ends at first tip entry, or at T if none occurs. Recovery begins at the command boundary ending that entry interval, or at T . At handover, C_{exit} penalizes weighted squares of command speed, positive upward velocity, and positive acceleration along the command velocity; the last term is zero at zero speed. Infeasible candidates receive $J = -\infty$.

Initial planning by sampling. An offline MPPI-inspired search uses exponential trajectory weighting [22] and maintains several proposal means. For each mean $\boldsymbol{\mu}$, it samples $\boldsymbol{\theta}^{(b)} = \boldsymbol{\mu} + \boldsymbol{\epsilon}^{(b)}$. The perturbations follow a fixed mixture of correlated Gaussian distributions at several scales. Their

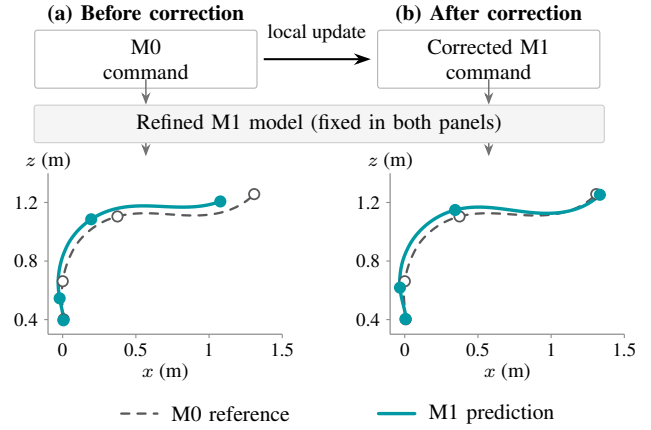


Fig. 2: M0-to-M1 command correction, shown in the world x - z plane. The fixed M1 model predicts motion under (a) the original M0 commands and (b) the corrected M1 commands. Dashed curves show the original M0 predicted tip reference; markers pair equal timestamps. All curves are saved simulation predictions.

covariance favors coherent variations of neighboring control points while exploring all free coordinates. Candidates are simulated in parallel.

Let $\mathcal{V}$ index a proposal's feasible random samples and $J_b = J(\boldsymbol{\theta}^{(b)})$. For nonempty $\mathcal{V}$,

$$w_b = \frac{\exp(J_b/\lambda)}{\sum_{a \in \mathcal{V}} \exp(J_a/\lambda)}, \quad \boldsymbol{\mu}^+ = \sum_{b \in \mathcal{V}} w_b \boldsymbol{\theta}^{(b)}. \quad (14)$$

Otherwise, the mean remains unchanged. The positive temperature λ is adjusted separately for each proposal to target a prescribed effective sample count, $(\sum_{b \in \mathcal{V}} w_b^2)^{-1}$. Unperturbed proposal means and the best command found so far are also evaluated; only random samples enter the weighted update. The highest-scoring feasible candidate is retained across iterations. Search ends at a score plateau or a fixed iteration budget.

Command correction with the refined model. Figure 2 shows the M0-to-M1 correction: the fixed M1 model predicts motion under the original M0 commands and the corrected M1 commands. The desired tip trajectory and timing remain unchanged. The initial MPPI plan's predicted tip and vehicle trajectories, $\mathbf{r}_{\text{tip}}^0(t)$ and $\mathbf{p}_q^0(t)$, and position commands $\mathbf{p}_d^0(t)$ form the fixed reference. Superscript 0 identifies this plan; ref denotes the earlier shape prior used only in initial planning.

Over the original plan's fixed handover interval $[0, T_h]$, where $0 < T_h \leq T$, we minimize

$$\begin{aligned} C_{\text{track}}(\boldsymbol{\theta}) &= w_{\text{tip}} \langle \|\mathbf{r}_{\text{tip}} - \mathbf{r}_{\text{tip}}^0\|_2^2 \rangle + w_q \langle \|\mathbf{p}_q - \mathbf{p}_q^0\|_2^2 \rangle \\ &\quad + w_d \langle \|\mathbf{p}_d - \mathbf{p}_d^0\|_2^2 \rangle. \end{aligned} \quad (15)$$

Unmarked states are predictions under the refined model. The brackets average squared distances at matching times over $[0, T_h]$, using uniform prediction samples for the first two terms and command samples for the third. Positive dimensionless weights prioritize tip tracking, with smaller

penalties for vehicle-path deviation and changes from the original commands. The reference, timing, objective, and admissibility constraints remain fixed across correction rounds.

Starting from the latest executed command, regularized Gauss–Newton updates adjust the spline controls affecting $[0, T_h]$. Finite differences estimate tip and vehicle sensitivities; command positions depend linearly on the controls. At each iteration, we linearize the tracking errors and solve a constrained local least-squares problem with bounded increments and quadratic damping in scaled control coordinates. Backtracking accepts only updates that reduce C_{track} in nonlinear simulation and satisfy the command and rollout checks. Iteration ends at small improvement, failure to find a feasible improving step, or the computation budget.

Measurements affect command correction through model refinement, described next. Both optimization stages verify the selected command in an independent rollout, including a recovery with continuous position, velocity, and acceleration.

C. Model Refinement from Physical Executions

The initial model $\mathcal{M}_0$ comes from preliminary calibration. Each update starts from the preceding model and fits vehicle-response parameters, cable coefficients, and learned acceleration corrections to execution measurements. The fitting objective compares predicted and measured motion and does not depend on the target or planned reference. Cable length, masses, attachment geometry, and integration settings remain fixed.

Recordings contain the executed commands, measured vehicle pose, and cable-marker positions. Equation (6) maps measured vehicle pose to attachment motion. Each fitting window starts from preceding measurements and advances without resetting predicted states to later observations. Missing observations are excluded from the loss.

The fitting set combines designated recordings from the latest batch, earlier whipping batches, and preliminary motions. Fixed group weights balance these conditions; within each group, recordings have equal weight distributed among eligible windows. Final test recordings remain outside fitting and model selection.

Vehicle fitting compares command-driven predictions with measured vehicle motion. Cable fitting uses the measured attachment trajectory, separating cable-dynamics errors from errors in predicted vehicle motion. Both use a weighted simulation-error loss,

$$\mathcal{L}_{\text{data}} = \sum_{o \in \mathcal{O}} w_o \rho \left(\frac{\|\mathbf{e}_o\|_2^2}{\sigma_o^2} \right), \quad (16)$$

where $\mathcal{O}$ contains valid observations, $\mathbf{e}_o \in \mathbb{R}^3$ is the prediction error, and $w_o \geq 0$ with $\sum_{o \in \mathcal{O}} w_o = 1$. Position errors are predicted minus measured positions. Orientation errors are the principal rotation vectors of $\mathbf{R}_o^\top \mathbf{R}_o^{\text{meas}}$, where $\mathbf{R}_o$ and $\mathbf{R}_o^{\text{meas}}$ are the predicted and measured orientations. Fixed positive scales σ_o normalize position and orientation errors. The robust penalty $\rho(z) = 2(\sqrt{1+z} - 1)$ limits the influence of large errors. Cable losses emphasize the tip while retaining the other observed non-attachment markers.

The update proceeds in four stages:

- 1) *Quadrotor response.* Fit $\mathbf{K}_p$, $\mathbf{K}_v$, $\mathbf{K}_a$, and τ_d using position errors, keeping the inherited vehicle correction fixed. Separately fit $\mathbf{S}$ and τ_R using orientation errors.
- 2) *Quadrotor correction.* Fix the nominal translation parameters and fit the vehicle network using position errors over simulated trajectories. Then refit $\mathbf{S}$ and τ_R : the corrected position and velocity predictions alter the nominal acceleration used to predict attitude.
- 3) *Cable physics.* Fit EI , C_b , and γ_c using measured attachment motion and marker-position errors, keeping the inherited cable correction fixed.
- 4) *Cable correction.* Fix the fitted cable coefficients and fit the cable network using the same measured attachment trajectory and marker loss.

Nominal parameter fits use fixed bounds and quadratic penalties on changes from the initial values of each fit. Positive parameters use logarithmic changes; the delay uses changes normalized by a fixed time scale.

The neural stages penalize correction magnitude and deviation from the inherited correction:

$$\mathcal{L}_{\text{NN}} = \mathcal{L}_{\text{data}} + \frac{\lambda_a}{3a_{\text{max}}^2} \left\langle \|\Delta \mathbf{a}\|_2^2 + \|\Delta \mathbf{a} - \Delta \mathbf{a}^{\text{prev}}\|_2^2 \right\rangle. \quad (17)$$

Here $\Delta \mathbf{a}$ denotes the vehicle or free-node cable correction, and $\Delta \mathbf{a}^{\text{prev}}$ is the inherited network evaluated at the same inputs. The brackets average over weighted fitting windows and simulated times, and additionally over free nodes for the cable. Cable averages exclude times without valid marker observations. The bound a_{max} normalizes acceleration, $\lambda_a \geq 0$ sets regularization strength, and 3 averages the components. Training differentiates through complete rollouts and requires no measured acceleration labels.

The retained $\mathcal{M}_0$ has no active cable correction. The first update adds a network initialized to zero output, with $\Delta \mathbf{a}^{\text{prev}} = \mathbf{0}$ in (17). This increases model capacity; later updates inherit the weights and preserve the architecture.

Each fitting stage retains its best numerically valid training iterate. Development recordings support model selection and diagnostic comparisons; the selected models are frozen before final evaluation. Complete command-to-tip prediction evaluates the assembled model and is distinct from cable prediction driven by measured attachment motion. It is not jointly optimized during the staged fits.

The updated predictor returns to command correction or subsequent MPPI planning. We evaluate prediction accuracy on held-out executions and assess command correction through physical reference tracking and target approach. Original commands and their preflight forecasts are preserved across model updates.

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